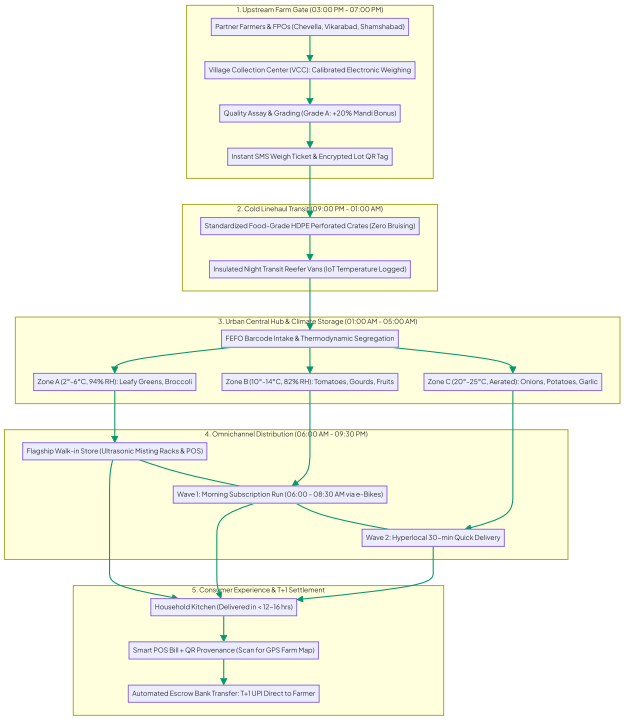
Jaya Raithu Market జయ రైతు మార్కెట్

Farm-to-Fork Direct Sourcing & Zero-Waste Omnichannel Agritech Model

Hub: Banjara Hills Store #01 (Telangana)
Cluster: 50–120 km Peri-Urban Belt

1. End-to-End Supply Chain Flow (Farm Gate to Dining Table in 12 Hours)

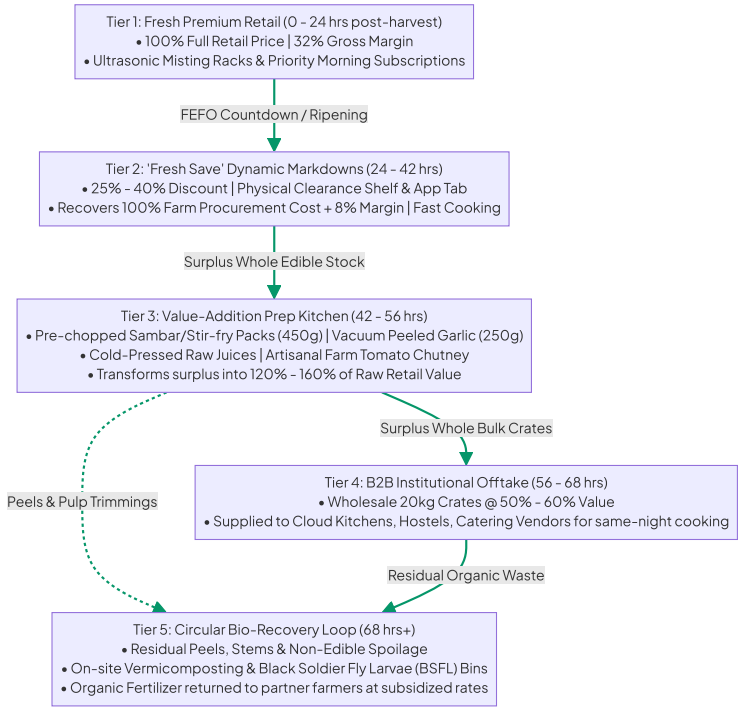
Eliminates 5 to 7 traditional APMC middlemen. Produce harvested in the evening arrives on urban retail misting racks by 5:00 AM, guaranteeing 60%+ farmer realization and peak leaf hydration.



Chain Stage	Operational Window	Core Action & Standards	Preservation / Quality Metric
Farm Harvest & VCC	03:00 PM – 07:00 PM	Electronic weighing, quality grading, perforated HDPE crate packing	Zero rough jute bag bruising; hydration locked
Night Reefer Linehaul	09:00 PM – 01:00 AM	Insulated transport within 50–120 km radius to Banjara Hills hub	Night transit minimizes metabolic heat build-up
Urban 3-Zone Storage	01:00 AM – 05:00 AM	Zone A (4°C, 94% RH), Zone B (11°C, 82% RH), Zone C (22°C Aerated)	Thermodynamic segregation stops yellowing
Delivery & Walk-in Store	06:00 AM – 08:30 AM	Electric bike batch milk-run & misting rack displays	**Delivered within 12–16 hrs of harvest**

2. The 5-Tier Zero-Waste Produce Cascade Flow (< 3% Net Spoilage)

To conquer perishable shrinkage, fresh produce moves through an automated algorithmic devaluation and repurposing funnel. Unsold items from Day 1 are discounted or converted into gourmet value-add kitchen packs.

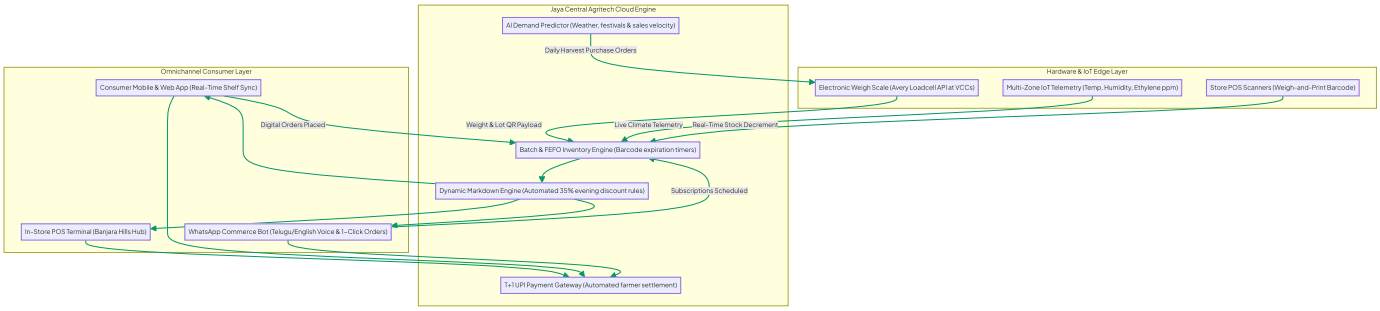


Cascade Tier	Lifecycle Window	Primary Channel / Action	Economic Recovery & Margin
Tier 1: Premium Retail	Day 0 – Day 1 (First 24 hrs)	Misting racks, morning milk-run baskets, 30m express delivery	**100% Retail Price (32% Gross Margin)**
Tier 2: "Fresh Save" Markdown	Day 1 PM – Day 2 AM (24–42 hrs)	Clearance shelf & dedicated app tab with 35% discount	**70%–80% Value (Recovers farm cost + 8%)**
Tier 3: Prep Kitchen Deli	Day 2 (42–56 hrs)	Pre-cut vegetable packs, peeled garlic, cold juices, chutney	**120%–160% Raw Value (High-Margin Value Add)**
Tier 4: B2B Offtake	Day 2 PM (56–68 hrs)	Bulk crates sold to cloud kitchens, hostel messes, tiffin centers	**50%–60% Value (Covers procurement & transit)**
Tier 5: Circular Recovery	Peels, stems, trim (68+ hrs)	Vermicomposting & Black Soldier Fly Larvae organic bio-fertilizer	**Closed loop: Soil nutrients back to farmers**

♻ **Zero Landfill Impact:** Produce waste sent to landfills is constrained to **under 1.9%**, converting potential loss into organic fertilizer and high-margin kitchen packs.

3. Technology Architecture & Real-Time Data Synchronization

Hardware edge devices at the farm gate connect seamlessly with cloud FEFO inventory algorithms and customer channels to automate stock rotation and direct farmer payouts.

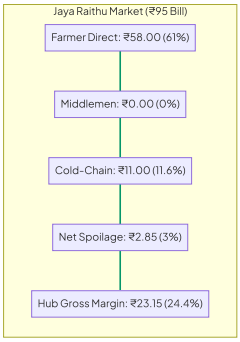


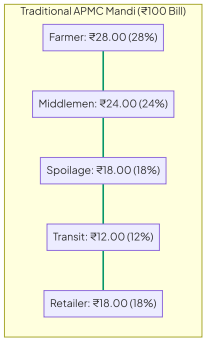
Software Component	Key Responsibility	Business Output
FEFO Inventory Engine	Tracks exact harvest hour per QR lot; alerts staff when lots hit 24h & 42h windows	Zero expired goods on flagship shelves
Dynamic Markdown Engine	Pushes algorithmic discounts to app & POS simultaneously at 06:30 PM	Rapid sell-through of day-1 produce
WhatsApp Commerce Bot	Enables 1-click morning basket subscription and regional voice ordering in Telugu	68% lower customer acquisition cost
T+1 Escrow Bank Service	Auto-reconciles POS sales and disburses bank payouts directly to farmer UPI VPA	Guaranteed next-morning farmer payment

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Consumer Rupee Value Distribution & Unit Economics

Farmer Realization: 61.0% (Mandi: 28.0%)
Middlemen Commissions: ₹0.00 (Eliminated)





Cost Component	Traditional Mandi Chain	Jaya Raithu Market	Economic Variance
Retail Price Paid by Customer	₹100.00	**₹95.00**	**₹5.00 Consumer Saving (5% Cheaper)**
Direct Farmer Realization	₹28.00 (28%)	**₹58.00 (61.0%)**	**+107% Higher Farmer Cash Inflow**
Middlemen / Commission Agents	₹24.00 (24%)	**₹0.00 (0%)**	100% Intermediary Elimination
Transit Spoilage / Shrinkage	₹18.00 (18%)	**₹2.85 (3.0%)**	84% Reduction in Produce Waste
Logistics & Cold-Chain Handling	₹12.00 (12%)	**₹11.00 (11.6%)**	Standardized reusable HDPE crates
Platform & Store Gross Profit	₹18.00 (18%)	**₹23.15 (24.4%)**	Healthy, Sustainable Unit Economics

Executive Summary: The Jaya Raithu Market model provides an equitable, high-margin, scalable agritech blueprint that pays farmers more, sells to consumers cheaper, and achieves near-zero waste.